Md. Meraj Mridha

Junior Software Engineer (Android) @ Deep Mind Labs · Full Stack & Mobile Developer

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SUMMARY

Builds a production healthcare Android app at Deep Mind Labs and full products across web, Android and backend (.NET, Node.js, Next.js, Kotlin, Flutter, PostgreSQL). Works test-first with CI/CD and uses Claude Code, agent skills and MCP for AI-assisted development.

EXPERIENCE

Junior Software Engineer (Android) — Deep Mind Labs / NPC Healthcare Oct 2025 – Present

DMHAI – Healthcare Patient Assessment App · Kotlin, Jetpack Compose, MVVM, Hilt, Retrofit, OkHttp, Room, CameraX, ML Kit, Media3

- Built a production Android app (v1.4.8, ~48K lines of Kotlin) for patient intake questionnaires, appointment-based assessments and guided range-of-motion (ROM) exercise recording for clinicians.
- Designed a multi-tenant networking layer: the app finds a patient's practice by email, then OkHttp interceptors switch the server address and add tenant and JWT headers on every request.
- Built real-time on-device pose detection (CameraX + ML Kit) with landmark smoothing, a three-frame visibility check and a countdown-and-record state machine; live joint-angle checks from server keypoints and multi-segment recording.
- Made the app reliable: JWT refresh on 401 (unit tested), fallback questionnaire API, timeout handling and a connection-quality banner; offline audio/GIF caching with Room and English/Spanish audio.
- Built a chatbot-style intake flow with a front/back body map for pain areas; managed releases (signing, R8/ProGuard, AAB).

Product Engineer — Independent SaaS products 2026 – Present

- Design and ship full products for the Bangladesh market (below) with bKash/Nagad/SSLCommerz payments, Bangla/English UIs, CI pipelines, Docker/Terraform deploys and ADR-driven design.

EDUCATION

Shahjalal University of Science & Technology — B.Sc. in Software Engineering 2022 – present · Sylhet

SKILLS

Languages: TypeScript, JavaScript, C#, Kotlin, Dart, Python, Java, C++, C, SQL

Backend: .NET / ASP.NET Core, EF Core, SignalR, Hangfire, Node.js, Express.js, NestJS, Prisma, Spring Boot, REST APIs

Frontend & Mobile: React.js, Next.js, Tailwind CSS, Ant Design, TanStack Query, Jetpack Compose, MVVM, Hilt, Retrofit/OkHttp, Room, CameraX, ML Kit, Media3, Flutter, Expo, Android (Java)

Databases: PostgreSQL (PostGIS, row-level security), MySQL, MongoDB, Redis, Firebase

DevOps & Cloud: Docker, GitHub Actions, Terraform, AWS EC2, Google Cloud, Vercel, Caddy, Git

AI & Claude: Claude Code, Claude API (Anthropic SDK), Agent Skills, MCP, CLAUDE.md workflows, prompt engineering, Google Gemini API, ML (NumPy, Pandas, SVM, LSTM)

Practices: Test-Driven Development, Clean Architecture, Modular Monolith, Multi-tenant SaaS, System Design & ADRs, OOP, Design Patterns, DSA, Selenium, JUnit

PROJECTS

FarmAI Platform — .NET 10, PostgreSQL, Next.js, Kotlin/Compose, Gemini, Hangfire 2026

- AI crop disease diagnosis (upload → background job → Gemini) that builds a basket of real, priced treatments; marketplace with vendors, cart and cash-on-delivery orders; farmer Android app with OTP sign-in. Test-driven with CI coverage gates.

SOS Emergency Bangladesh — .NET 8, SignalR, PostgreSQL + PostGIS, Next.js 14, Kotlin 2026

- One-tap SOS platform connecting citizens with mechanics, ambulances and fire services: GPS nearby search, live location and chat, QR vehicle stickers, OTP + JWT auth, bilingual web app (46 routes) and Android app.

Turf Booking SaaS — .NET 10, PostgreSQL, Redis, Next.js, Expo 2026

- Multi-tenant booking platform: tenant isolation with EF Core filters + row-level security, a DB exclusion constraint against double-booking, per-turf payment merchants, transactional outbox.

Also: Sebika — Bangla-first AI prescription reader (Kotlin, Node.js, PostgreSQL, Terraform) · **Taka Tracker** — Flutter + NestJS money tracker with subscriptions · ICERIE website (Next.js, team project) · SmartSchool · Library App · SkillMate AI · FarmAI v1

COMPETITIONS & RESEARCH

- BdSL Bangla Sign Language — feature extraction, SVM/LSTM models
- Astrobee Kibo Robot Programming Challenge (Java)
- UIHP Farm AI Project — 2nd Runner-up
- Therap Javafest 2025 — Qualified participant
- EDGE Data Science — Certified trainee

LEADERSHIP

- **Vice President**, Chokh Film Society (CFS) *Aug 2025 – present*
- **General Secretary**, Tangail Students' Association of SUST *May 2023 – Nov 2024*
- **Campus Ambassador**, University Innovation Hub Program *Nov 2022 – present*

PROBLEM SOLVING

400+ problems solved in C/C++ on LeetCode, Codeforces, CodeChef and VJudge.